

Zachary Silliman

zacharysilliman@gmail.com | NYC | [Portfolio](#)

Software Engineer (5+ Years) specialized in real-time graphics and ML-driven experiences

WORK EXPERIENCE

Framestore

April 2021 – Present

Senior Immersive Developer | April 2025 - Present

New York City, NY

Worked on real-time Unreal Engine tooling and ML-driven interactive characters.

Brand Character Animation Tools ([sample](#))

- Set up Xsens/Faceware motion capture pipelines for character animation
- Trained a real-time PyTorch head-animation model, deployed via Unreal's NNE plugin
- Built Unreal editor tools for creating/editing lip-sync assets with NVIDIA Audio2Face
- Designed an animation system for character equipment and props
- Built a React app to control a pixel-streamed real-time character on a timeline

Immersive Developer | April 2021 - April 2025

Prior to my promotion, I worked on an assortment of realtime interactive installations.

Cadillac, Art of You (Immersive Experience, [link](#))

- Drove a real-time Niagara particle system with live speech-to-text input, letting users sculpt a bespoke Celestiq car on an 18×18 ft display (UE5)
- Adapted nanoVDB plugin to frame-interpolate baked Houdini fluid sims for real-time playback

One Vanderbilt, Summit Unity (Immersive Experience, [link](#))

- Optimized a real-time, 6-PC nDisplay render cluster for a 47.5×12 ft LED wall, using Unreal Insights to resolve sync/performance issues
- Built the Node.js/MongoDB/Vue.js backend for live asset tracking and remote debugging of an ongoing installation

Japan Research Institute

June 2019 – September 2019

Information Security Practitioner

New York City, NY

Managed risks, vulnerabilities, identity, access and a variety of enterprise security tools

- SANS-trained in incident response and penetration testing; earned GSEC and GCFA certifications

Projects

Indie Game, Null Transit

December 2025

Released 2025

New York City, NY

- Designed, implemented and marketed my own 2D action-platformer called *Null Transit*.
- Available [here](#) on Steam

SKILLS

- **Languages:** C++, C#, Python, Typescript, Javascript, Rust, PowerShell, Batch
- **Frameworks:** PyTorch, Node.js, React, Django, Axum, Jekyll
- **Tools:** Unreal Engine, Unity, Blender, Xsens, Manus, Faceware, Jenkins, Perforce, Git, Jira, P4 Plan
- **For Fun:** Rollerblading, rock climbing, philosophy

EDUCATION

Cornell University

Sep 2015 – May 2019

Bachelor of Arts, **Computer Science** specialization in **Mathematics**

Ithaca, NY